

Math 237A Midterm

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1 Setup:

Let $\mathbb{G} = \mathbb{G}(2, 4)$ be the Grassmannian variety parametrizing 2-dimensional subspaces in a 4-dimensional vector space, or equivalently, parametrizing $\mathbb{P}^1$ in a given $\mathbb{P}^3$. Recall that the Plücker embedding identifies $\mathbb{G}$ with a quadric hypersurface in $\mathbb{P}^5$.

2 Notations & Tools:

Let k^4 be the standard 4-dimensional k -vector space. If view $\mathbb{P}^3 := (k^4 \setminus \{0\}) / \sim$, where any nonzero vector $v, u \in k^4$, $v \sim u$ iff $u = \lambda v$ for some $\lambda \in k$, then it defines a quotient map $\pi : k^4 / \{0\} \twoheadrightarrow \mathbb{P}^3$. Here are some things we're assuming:

(1) Line $\ell \subset \mathbb{P}^3$ as 2-dimensional subspace $\tilde{\ell} \subset k^4$:

For each line $\ell \subset \mathbb{P}^3$, there exists $p, q \in \mathbb{P}^3$ (where $p \neq q$) such that $\ell = \{sp + tq \mid [s, t] \in \mathbb{P}^1\}$ (where given $p = [p_0, p_1, p_2, p_3], q = [q_0, q_1, q_2, q_3]$, $sp + tq = [sp_i + tq_i]_{0 \leq i \leq 3} \in \mathbb{P}^3$). Then, notice that let $p' = (p_0, p_1, p_2, p_3)$ and $q' = (q_0, q_1, q_2, q_3)$ in k^4 , one has $\pi^{-1}(\ell) \cup \{0\} = \text{span}\{p', q'\}$:

Since any $\pi(p') = p$, $\pi(q') = q$, with $p \neq q$, this shows that p, q are not scalar multiples of each other, hence linearly independent. Then, for any nonzero $(s, t) \in \mathbb{A}^2$ (correspond to a point $[s, t] \in \mathbb{P}^1$), where at least one of s, t is nonzero, then $sp + tq \neq 0$ due to linear independence. Hence, one has $\pi(sp' + tq') = [sp_i + tq_i]_{0 \leq i \leq 3} \in \ell$, showing that $\pi(\text{span}\{p', q'\} \setminus \{0\}) = \ell$. Hence, WLOG can say ℓ is identified with a 2-dimensional subspace (here it's $\text{span}\{p', q'\}$) in k^4 , which we'll denote as $\tilde{\ell} \subseteq k^4$.

(2) Plane $H \subset \mathbb{P}^3$ as 3-dimensional subspace $\tilde{H} \subset k^4$:

Using similar logic, any plane $H \subset \mathbb{P}^3$ that has distinct points $p, q, w \in \mathbb{P}^3$ satisfying $H = \{sp + tq + rw \mid [s, t, r] \in \mathbb{P}^2\}$, one has $\pi^{-1}(H) \cup \{0\} = \text{span}\{p', q', w'\}$ (where $p', q', w' \in k^4$ are representatives of $p, q, w \in \mathbb{P}^3$ under quotient), hence WLOG can also say H is identified with a 3-dimensional subspace in k^4 , we'll denote as $\tilde{H} \subseteq k^4$.

(3) Plücker Embedding with Change of Basis:

Fixing a basis $e_0, e_1, e_2, e_3 \in k^4$, then any line $\ell \subset \mathbb{P}^3$ (or 2D subspace $\tilde{\ell} \subset k^4$), if $\tilde{\ell} = \text{span}\{p, q\}$ (where $p = (p_0, p_1, p_2, p_3)$ and $q = (q_0, q_1, q_2, q_3)$), then the Plücker Embedding $P : \mathbb{G} \rightarrow \mathbb{P}^5$ can be defined as:

$$\ell := \begin{pmatrix} p_0 & p_1 & p_2 & p_3 \\ q_0 & q_1 & q_2 & q_3 \end{pmatrix}, \quad P(\ell) = [X_{01}, X_{02}, X_{03}, X_{12}, X_{13}, X_{23}], \quad X_{ij} := \det \begin{pmatrix} p_i & p_j \\ q_i & q_j \end{pmatrix} \quad (2.1)$$

Which, given any $g \in \text{GL}_2(k)$, the $g \cdot \begin{pmatrix} p_0 & p_1 & p_2 & p_3 \\ q_0 & q_1 & q_2 & q_3 \end{pmatrix}$ is invariant under Plücker Embedding (since then the new $X_{ij} = \det(g) \cdot \det \begin{pmatrix} p_i & p_j \\ q_i & q_j \end{pmatrix}$), hence it's well-defined under change of basis in $\tilde{\ell}$.

Furthermore, if we choose another basis $f_0, f_1, f_2, f_3 \in k^4$, and consider $g \in \text{GL}_4(k)$ that serves as a change-of-basis matrix from $\{f_1, f_2, f_3, f_4\}$ to $\{e_1, e_2, e_3, e_4\}$ (as column vectors). Then, given a line $\ell \subset \mathbb{P}^3$, $\ell = \text{span}\{p, q\}$ (where $p, q \in k^4$ are linearly independent). Let $p = (p_i)_{0 \leq i \leq 3}$ under basis e_i , and $p = (p'_i)_{0 \leq i \leq 3}$ under basis f_i (use the same notation for q). Let ℓ_e, ℓ_f denotes the 2×4 matrix representation of ℓ under basis e_i and f_i respectively. Then, together with g , they satisfy the following:

$$g = \begin{pmatrix} a_{00} & a_{01} & a_{02} & a_{03} \\ a_{10} & a_{11} & a_{12} & a_{13} \\ a_{20} & a_{21} & a_{22} & a_{23} \\ a_{30} & a_{31} & a_{32} & a_{33} \end{pmatrix} \in \text{GL}_4(k) \quad (2.2)$$

$$\ell_e \cdot g = \ell_f \implies \begin{pmatrix} p_0 & p_1 & p_2 & p_3 \\ q_0 & q_1 & q_2 & q_3 \end{pmatrix} \begin{pmatrix} a_{00} & a_{01} & a_{02} & a_{03} \\ a_{10} & a_{11} & a_{12} & a_{13} \\ a_{20} & a_{21} & a_{22} & a_{23} \\ a_{30} & a_{31} & a_{32} & a_{33} \end{pmatrix} = \begin{pmatrix} p'_0 & p'_1 & p'_2 & p'_3 \\ q'_0 & q'_1 & q'_2 & q'_3 \end{pmatrix} \quad (2.3)$$

Now, let $[X_{ij}]_{0 \leq i < j \leq 3} = P(\ell_e)$ and $[X'_{ij}]_{0 \leq i < j \leq 3} = P(\ell_f)$, we can in fact show that $P(\ell_f)$ and $P(\ell_e)$ are related using a linear transformation: For definiteness, we'll demonstrate only the relation on X'_{01} . Using the above formula, one can show the following:

$$X_{01} = \det \begin{pmatrix} p_0 & p_1 \\ q_0 & q_1 \end{pmatrix}, \quad X'_{01} = \det \begin{pmatrix} p'_0 & p'_1 \\ q'_0 & q'_1 \end{pmatrix} = \det \begin{pmatrix} \sum_{i=0}^3 a_{i0} p_i & \sum_{j=0}^3 a_{j1} p_j \\ \sum_{i=0}^3 a_{i0} q_i & \sum_{j=0}^3 a_{j1} q_j \end{pmatrix} \quad (2.4)$$

(Note: The equation for X'_{01} is given by change of basis formula).

Which, using the multilinearity of determinant (that can break down summation in each row/column, and factor out scalars in each row/column), we get the following:

$$X'_{01} = \det \begin{pmatrix} \sum_{i=0}^3 a_{i0} p_i & \sum_{j=0}^3 a_{j1} p_j \\ \sum_{i=0}^3 a_{i0} q_i & \sum_{j=0}^3 a_{j1} q_j \end{pmatrix} = \sum_{i=0}^3 \sum_{j=0}^3 a_{i0} a_{j1} \det \begin{pmatrix} p_i & p_j \\ q_i & q_j \end{pmatrix} = \sum_{i=0}^3 \sum_{j=0}^3 a_{i0} a_{j1} X_{ij} \quad (2.5)$$

Hence, applying similar logic one can show that each entry of $P(\ell_f) = [X'_{ij}]_{1 \leq i < j \leq 3}$ can be obtained using a linear transformation (with coefficients from matrix g), and the entries of $P(\ell_e) = [X_{ij}]_{0 \leq i < j \leq 3}$. Also, notice that such linear transformation is invertible (at least on $\mathbb{G} \subset \mathbb{P}^5$), due to the fact that g is invertible, and $\ell_e = \ell_f \cdot g^{-1}$ (recall that because of change of basis, $\ell_f = \ell_e \cdot g$), hence the whole process is invertible.

As a result, given the change of basis matrix $g \in \text{GL}_4(k)$ from f_i to e_i , there exists a linear transformation $T_g : \mathbb{P}^5 \rightarrow \mathbb{P}^5$ (that's invertible on embedding of $\mathbb{G} \subset \mathbb{P}^5$), such that the following commutes:

$$\begin{array}{ccc} \mathbb{G} & \xrightarrow{P} & \mathbb{P}^5 \\ \downarrow \scriptstyle - \cdot g & & \downarrow \scriptstyle T_g \\ \mathbb{G} & \xrightarrow{P} & \mathbb{P}^5 \end{array}$$

Where $- \cdot g$ denotes a change of representation $\ell_e \mapsto \ell_f$, and T_g represents the linear transformation $P(\ell_e) \mapsto P(\ell_f)$. Also, the top $\mathbb{G}$ uses e_i , while the bottom $\mathbb{G}$ uses f_i as matrix representation. **So, change of basis on k^4 just results in a different embedding of $\mathbb{G} \subset \mathbb{P}^5$, and such embeddings are associated by linear transformations (invertible on the images of embeddings), hence it preserves lines and planes in $\mathbb{G}$.**

So, for each part of the question, we'll most likely perform a change of basis for ease of computation.

3 D

Problem 1

For any point $p \in \mathbb{P}^3$ and plane $H \subset \mathbb{P}^3$ containing p , let $\Sigma_{p,H} \subset \mathbb{G}$ be the locus of lines in $\mathbb{P}^3$ passing through p and lying in H . Show that under the Plücker embedding $\mathbb{G} \rightarrow \mathbb{P}^5$, $\Sigma_{p,H}$ is carried to a line, and that conversely every line in $\mathbb{P}^5$ lying on $\mathbb{G}$ is of the form $\Sigma_{p,H}$ for some p and H .

Solution:

Σ_p embeds to a line in $\mathbb{P}^5$:

If identify $p \in k^4$ as a representative of $p \in \mathbb{P}^3$, then as a 3-dimensional subspace, $\tilde{H} \subset k^4$ contains p (so under projection $p \in H \subset \mathbb{P}^3$, the corresponding plane of $\tilde{H} \subset k^4$). Since $p \neq 0$, extend it to a basis p, h_1, h_2 of $\tilde{H}$, then further extend it to a basis p, h_1, h_2, v of k^4 . We'll use this basis for Plücker Embedding.

For every line $\ell \subset \Sigma_{p,H}$ satisfying $p \in \ell \subset H$ (passes through p , lies in H), since $\tilde{\ell} \subset \tilde{H}$ is a 2-dimensional subspace containing p , there exists $q \in \tilde{H}$ such that $\tilde{\ell} = \text{span}\{p, q\}$ (where p, q are linearly independent). Then, since $q \in \tilde{H}$, there exists unique $(s, t, r) \in \mathbb{A}^3$ such that $q = sp + th_1 + rh_2$; and based on the linear independence of p, q , one must have t or r being nonzero (if $t = r = 0$, $q = sp$ cannot be linearly independent with p). Hence, $\tilde{\ell} = \text{span}\{p, sp + th_1 + rh_2\} = \text{span}\{p, th_1 + rh_2\}$ (with t or r being nonzero), showing that one can pick $[t, r] \in \mathbb{P}^1$ such that $\tilde{\ell} = \text{span}\{p, th_1 + rh_2\}$.

Now, since under the ordered basis $\{p, h_1, h_2, v\}$ of k^4 , $p = (1, 0, 0, 0)$ and $th_1 + rh_2 = (0, t, r, 0)$, hence under the chosen basis of k^4 , the line $\ell \subset \mathbb{P}^3$ can be represented the following matrix:

$$\ell = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & t & r & 0 \end{pmatrix} \quad (3.1)$$

So, under Plücker Embedding, one yields:

$$P(\ell) = [\det \begin{pmatrix} 1 & 0 \\ 0 & t \end{pmatrix}, \det \begin{pmatrix} 1 & 0 \\ 0 & r \end{pmatrix}, \det \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \det \begin{pmatrix} 0 & 0 \\ t & r \end{pmatrix}, \det \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \det \begin{pmatrix} 0 & 0 \\ r & 0 \end{pmatrix}] = [t, r, 0, 0, 0, 0] \quad (3.2)$$

Hence, consider the line embedding $\varphi : \mathbb{P}^1 \rightarrow \mathbb{P}^5$ by $\varphi([t, r]) = [t, r, 0, 0, 0, 0]$, then since each $\ell = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & t & r & 0 \end{pmatrix}$ has $P(\ell) = [t, r, 0, 0, 0, 0]$ (after choosing the basis), then one can see that $P(\Sigma_{p,H}) = \varphi(\mathbb{P}^1)$, showing that $\Sigma_{p,H}$ has its image being a line in $\mathbb{P}^5$.

Converse:

To prove the converse, suppose a line $\ell' \cong \mathbb{P}^1 \subset P(\mathbb{G}) \subset \mathbb{P}^5$ is parametrized by distinct $p, q \in P(\mathbb{G})$, and $\ell' = \{tp + rq \mid [t, r] \in \mathbb{P}^1\}$. Which, let $\tilde{\ell}_p, \tilde{\ell}_q \subset k^4$ be distinct 2-dimensional subspaces corresponding to points $p, q \in \ell'$ (so $p = P(\ell_p)$ and $q = P(\ell_q)$ under Plücker Embedding), we claim that $\tilde{\ell}_p \cap \tilde{\ell}_q \neq 0$:

Suppose the contrary that $\tilde{\ell}_p \cap \tilde{\ell}_q = 0$, they form a direct sum, then $\tilde{\ell}_p \oplus \tilde{\ell}_q = k^4$ (since $\dim_k(\tilde{\ell}_p \oplus \tilde{\ell}_q) = \dim_k(\tilde{\ell}_p) + \dim_k(\tilde{\ell}_q) = 4$). So, based on the ability of performing Change of Basis on k^4 , and resulting in another embedding of $\mathbb{G}$ into $\mathbb{P}^5$ that preserves lines and planes (this is proved in **2 Notations & Tools** part (3)), WLOG one can choose basis $e_0, e_1 \in \tilde{\ell}_p$, and $e_2, e_3 \in \tilde{\ell}_q$, which forms a basis e_0, e_1, e_2, e_3 of k^4 .

If using this ordered basis as matrix representation, then notice that $\tilde{\ell}_p = \text{span}\{e_0, e_1\}$ and $\tilde{\ell}_q = \text{span}\{e_2, e_3\}$ (or $\ell_p, \ell_q \in \mathbb{G}$) are represented as follow:

$$\ell_p = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \quad \ell_q = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (3.3)$$

Under Plücker Embedding, the two yields the following coordinates in $\mathbb{P}^5$:

$$p = P(\ell_p) = [1, 0, 0, 0, 0, 0], \quad q = P(\ell_q) = [0, 0, 0, 0, 0, 1] \quad (3.4)$$

Hence, under this new embedding, the line $\ell' = \{tp + rq \mid [t, r] \in \mathbb{P}^1\} = \{[t, 0, 0, 0, 0, r]\}_{[t, r] \in \mathbb{P}^1}$.

Since this line is assumed to be in the image of Plücker Embedding, then every $[t, r] \in \mathbb{P}^1$ should have $[t, 0, 0, 0, 0, r] \in P(\mathbb{G})$. For $[t, r] = [0, 1]$, we have the point $q \in P(\mathbb{G})$, which yields no problem (similarly for $[t, r] = [1, 0]$ that yields $p \in P(\mathbb{G})$).

Yet, when consider $t, r \neq 0$ (WLOG, say $t = 1$), since it corresponds to $[1, 0, 0, 0, 0, r] \in P(\mathbb{G})$, suppose it corresponds to a 2-dimensional subspace $\tilde{\ell} = \text{span}\{a, b\}$ (where $a = (a_i), b = (b_i)$ under the given basis $e_0, e_1, e_2, e_3 \in k^4$), so the line $\ell \subset \mathbb{G}$ is given under the following representation:

$$\ell = \begin{pmatrix} a_0 & a_1 & a_2 & a_3 \\ b_0 & b_1 & b_2 & b_3 \end{pmatrix} \quad (3.5)$$

then since for $[1, 0, 0, 0, 0, r] = P(\ell) = [X_{ij}]_{0 \leq i < j \leq 3}$, one has $1 = X_{01} = \det \begin{pmatrix} a_0 & a_1 \\ b_0 & b_1 \end{pmatrix}$, then up to a change of basis using matrix in $\text{GL}_2(k)$, WLOG one can assume $\begin{pmatrix} a_0 & a_1 \\ b_0 & b_1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, so ℓ is given as follow:

$$\ell = \begin{pmatrix} 1 & 0 & a_2 & a_3 \\ 0 & 1 & b_2 & b_3 \end{pmatrix} \quad (3.6)$$

Hence, we get that $[1, 0, 0, 0, 0, r] = P(\ell) = [1, b_2, b_3, -a_2, -a_3, a_2b_3 - a_3b_2]$, hence deduces that $a_2, a_3, b_2, b_3 = 0$, showing that $r = 0$. Yet, this contradicts our assumption that $r \neq 0$. So, the initial assumption must be false, showing that $\tilde{\ell}_p \cap \tilde{\ell}_q \neq \emptyset$.

As a result, let the nonzero vector $x \in \tilde{\ell}_p \cap \tilde{\ell}_q$, since $\tilde{\ell}_p \neq \tilde{\ell}_q$, one can choose nonzero vectors $x_p \in \tilde{\ell}_p$ and $x_q \in \tilde{\ell}_q$, such that $\tilde{\ell}_p = \text{span}\{x, x_p\}$ and $\tilde{\ell}_q = \text{span}\{x, x_q\}$, and x, x_p, x_q are linearly independent.

Finally, consider the 3-dimensional subspace $\tilde{H} = \text{span}\{x, x_p, x_q\}$ (where $H \subset \mathbb{P}^3$ is a plane). Under the basis $e_0, e_1, e_2, e_3 \in k^4$, let $x = (x_i)$, $x_p = (x_{p,i})$, and $x_q = (x_{q,i})$ be the coordinates, then $\ell_p, \ell_q \in \mathbb{G}$ are represented using the following matrices:

$$\ell_p = \begin{pmatrix} x_0 & x_1 & x_2 & x_3 \\ x_{p,0} & x_{p,1} & x_{p,2} & x_{p,3} \end{pmatrix}, \quad \ell_q = \begin{pmatrix} x_0 & x_1 & x_2 & x_3 \\ x_{q,0} & x_{q,1} & x_{q,2} & x_{q,3} \end{pmatrix} \quad (3.7)$$

Which, for all line $L \in \Sigma_{x,H}$ satisfying $x \in L \subset H$ (or 2-dimensional subspace $\tilde{L}$ satisfying $x \in \tilde{L} \subset \tilde{H}$), since $\tilde{L} = \text{span}\{x, tx_p + rx_q\}$ for some $[t, r] \in \mathbb{P}^1$ (due to the fact that $\tilde{H} = \text{span}\{x, x_p, x_q\}$), then one can choose another basis vector of $\tilde{L}$ as $sx + tx_p + rx_q$, where t or r is nonzero; hence $\tilde{L} = \text{span}\{x, sx + tx_p + rx_q\} = \text{span}\{x, tx_p + rx_q\}$, then $\tilde{L} \in \Sigma_{x,H}$ has the following matrix representative:

$$L = \begin{pmatrix} x_0 & x_1 & x_2 & x_3 \\ t(x_{p,0} + rx_{q,0}) & t(x_{p,1} + rx_{q,1}) & t(x_{p,2} + rx_{q,2}) & t(x_{p,3} + rx_{q,3}) \end{pmatrix} \quad (3.8)$$

Hence, $p(L) = [X_{ij}]_{0 \leq i < j \leq 3}$, where each $X_{ij} = \det \begin{pmatrix} x_i & x_j \\ t(x_{p,i} + rx_{q,i}) & t(x_{p,j} + rx_{q,j}) \end{pmatrix} = t \det \begin{pmatrix} x_i & x_j \\ x_{p,i} & x_{p,j} \end{pmatrix} + r \det \begin{pmatrix} x_i & x_j \\ x_{q,i} & x_{q,j} \end{pmatrix}$ (notice that the two determinants are coming from the (i, j) entry of $P(\ell_p)$ and $P(\ell_q)$ respectively), so $P(L) = tP(\ell_p) + rP(\ell_q) = tp + rq$.

Which, this shows that $\ell' = \{tp + rq \mid [t, r] \in \mathbb{P}^1\} = P(\Sigma_{x,H})$ (since there exists $L \in \Sigma_{x,H}$ such that $P(L) = tp + rq$ for all $[t, r] \in \mathbb{P}^1$, while every $L \in \Sigma_{x,H}$ is also in such form). Hence, we can

conclude that each line $\ell' \in \mathbb{G} \subset \mathbb{P}^5$ is of the form $\Sigma_{p,H}$ for some point p and plane $H \subset \mathbb{P}^3$, such that $p \in H$.

4 D

Problem 2

For any point $p \in \mathbb{P}^3$, let $\Sigma_p \subset \mathbb{G}$ be the locus of lines in $\mathbb{P}^3$ passing through p ; for any plane $H \subset \mathbb{P}^3$, let $\Sigma_H \subset \mathbb{G}$ be the locus of lines in $\mathbb{P}^3$ lying in H . Show that under Plücker Embedding, both Σ_p and Σ_H are carried into planes in $\mathbb{P}^5$, and that conversely any plane $\Lambda \cong \mathbb{P}^2 \subset \mathbb{G} \subset \mathbb{P}^5$ is either equal to Σ_p for some p or to Σ_H for some H .

Solution:

1. Case of Σ_p for $p \in \mathbb{P}^3$:

Let nonzero vector $p \in k^4$ be a representative of $p \in \mathbb{P}^3$ (abuse of notation), extend this to a basis p, e_1, e_2, e_3 of k^4 . We'll use this as a basis for Plücker Embedding.

Given any line $\ell \in \Sigma_p$ (or 2-dimensional subspace $\tilde{\ell} \in k^4$ with $p \in \tilde{\ell}$), extend $p \in \tilde{\ell}$ to a basis p, q of $\tilde{\ell}$, then since $q = ap + se_1 + te_2 + re_3$ for a unique $(a, s, t, r) \in \mathbb{A}^4$ (and one of s, t, r is nonzero, due to linear independence of the list p, q ; so $[s, t, r] \in \mathbb{P}^2$), then we get that $\tilde{\ell} = \text{span}\{p, q\} = \text{span}\{p, ap + se_1 + te_2 + re_3\} = \text{span}\{p, se_1 + te_2 + re_3\}$. Hence, under ordered basis $p, e_1, e_2, e_3 \in k^4$, the matrix representative of ℓ is as follow:

$$\ell = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & s & t & r \end{pmatrix} \quad (4.1)$$

And, $[s, t, r] \in \mathbb{P}^2$ can in fact be arbitrary. Hence, under Plücker Embedding, $\ell \in \Sigma_p$ yields:

$$P(\ell) = [s, t, r, 0, 0, 0] \in \mathbb{P}^5 \quad (4.2)$$

Hence, if consider the plane embedding $\varphi : \mathbb{P}^2 \rightarrow \mathbb{P}^5$ by $\varphi([s, t, r]) = [s, t, r, 0, 0, 0]$, the above shows that $P(\Sigma_p) = \varphi(\mathbb{P}^2)$, so Σ_p gets embedded into a plane in $\mathbb{P}^5$.

2. Case of Σ_H for plane $H \subset \mathbb{P}^3$:

Given $H \subset \mathbb{P}^3$ as a plane, $\tilde{H} \subset k^4$ is a 3-dimensional subspace. Fix basis $h_1, h_2, h_3 \in \tilde{H}$, and extend it to a basis $p, h_1, h_2, h_3 \in k^4$, we'll use this basis for Plücker Embedding.

Notice that given any line $\ell \in \Sigma_H$, or $\ell \subset H$ (which indicates that the 2-dimensional subspace $\tilde{\ell} \subset \tilde{H}$), we have $\tilde{\ell} = \text{span}\{a, b\} \subset \tilde{H}$. So, under the basis p, h_1, h_2, h_3 of k^4 , $a = (0, a_1, a_2, a_3)$ and $b = (0, b_1, b_2, b_3)$ (since $a, b \in \tilde{H} = \text{span}\{h_1, h_2, h_3\}$). Which, under Plücker Embedding, ℓ gets carried to the following (based on the matrix representative):

$$\ell = \begin{pmatrix} 0 & a_1 & a_2 & a_3 \\ 0 & b_1 & b_2 & b_3 \end{pmatrix}, \quad P(\ell) = \left[0, 0, 0, \det \begin{pmatrix} a_1 & a_2 \\ b_1 & b_2 \end{pmatrix}, \det \begin{pmatrix} a_1 & a_3 \\ b_1 & b_3 \end{pmatrix}, \det \begin{pmatrix} a_2 & a_3 \\ b_2 & b_3 \end{pmatrix} \right] \quad (4.3)$$

So, if consider the plane embedding $\varphi : \mathbb{P}^2 \rightarrow \mathbb{P}^5$ by $\varphi([s, t, r]) = [0, 0, 0, s, t, r]$, this shows that $P(\Sigma_H) \subseteq \varphi(\mathbb{P}^2)$, which is contained in a plane. To prove that $P(\Sigma_H) = \varphi(\mathbb{P}^2)$, given any $[0, 0, 0, s, t, r] \in \varphi(\mathbb{P}^2)$, there are two cases to consider:

- First, if $t \neq 0$ (WLOG, say $t = 1$), then if $\ell \in \Sigma_H$ above has $P(\ell) = [0, 0, 0, s, t, r]$, then one has $\det \begin{pmatrix} a_1 & a_3 \\ b_1 & b_3 \end{pmatrix} = t = 1$, so up to a change of basis matrix in $\text{GL}_2(k)$, one can assume $\begin{pmatrix} a_1 & a_3 \\ b_1 & b_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. Hence, one yields the following:

$$\ell = \begin{pmatrix} 0 & 1 & a_2 & 0 \\ 0 & 0 & b_2 & 1 \end{pmatrix}, \quad [0, 0, 0, s, t, r] = P(\ell) = [0, 0, 0, b_2, 1, a_2] \quad (4.4)$$

Hence, the line $\ell = \begin{pmatrix} 0 & 1 & r & 0 \\ 0 & 0 & s & 1 \end{pmatrix}$ satisfies $P(\ell) = [0, 0, 0, s, t, r]$ (if $t \neq 0$).

- Else, if $t = 0$, then if $\ell \in \Sigma_H$ above has $P(\ell) = [0, 0, 0, s, t, r]$, then one has $\det \begin{pmatrix} a_1 & a_3 \\ b_1 & b_3 \end{pmatrix} = t = 0$, showing $(a_1, a_3), (b_1, b_3)$ are linearly dependent. WLOG, up to some change of basis, one can say $\begin{pmatrix} a_1 & a_3 \\ b_1 & b_3 \end{pmatrix} = \begin{pmatrix} a_1 & a_3 \\ 0 & 0 \end{pmatrix}$ (subtracting rows, or swap rows). Hence, one yields the following:

$$\ell = \begin{pmatrix} 0 & a_1 & a_2 & a_3 \\ 0 & 0 & b_2 & 0 \end{pmatrix}, \quad [0, 0, 0, s, t, r] = P(\ell) = [0, 0, 0, a_1 b_2, 0, -a_3 b_2] \quad (4.5)$$

Since it needs to satisfy $s = a_1 b_2$ and $r = -a_3 b_2$, and since one of s, r must be nonzero (in case for $[s, t, r] = [s, 0, r] \in \mathbb{P}^2$), then one must have $b_2 \neq 0$ (if $b_2 = 0$, $s = a_1 b_2 = 0$ and $r = -a_3 b_2 = 0$ reaches a contradiction). So, WLOG, assume $b_2 = 1$ (by a scalar multiplication). Then, one yields $s = a_1$ and $r = -a_3$, so $\ell = \begin{pmatrix} 0 & s & a_2 & -r \\ 0 & 0 & 1 & 0 \end{pmatrix}$ satisfies $P(\ell) = [0, 0, 0, s, t, r]$ (where $t = 0$).

So, under both cases one has $[0, 0, 0, s, t, r] \in P(\Sigma_H)$, showing that $\varphi(\mathbb{P}^2) = P(\Sigma_H)$. Hence, Σ_H also gets carried to a plane in $\mathbb{P}^5$.

3. Converse:

Given a plane $\Lambda \cong \mathbb{P}^2 \subset \mathbb{G} \subset \mathbb{P}^5$, to prove that Λ is either Σ_p or Σ_H for some $p \in \mathbb{P}^3$, or plane $H \subset \mathbb{P}^3$, suppose an easier case where $\Lambda \neq \Sigma_H$ for any plane $H \subset \mathbb{P}^3$.

Because $\Lambda \cong \mathbb{P}^2$, there exists three distinct points $p, q, w \in \Lambda$, such that $\Lambda = \{sp + tq + rw \mid [s, t, r] \in \mathbb{P}^2\}$. Now, let $\tilde{\ell}_p, \tilde{\ell}_q, \tilde{\ell}_w \subset k^4$ be the corresponding 2-dimensional subspaces of p, q, w as points in $\mathbb{G}$, we claim that $\tilde{\ell}_p \cap \tilde{\ell}_q \cap \tilde{\ell}_w \neq 0$:

Suppose the contrary that $\tilde{\ell}_p \cap \tilde{\ell}_q \cap \tilde{\ell}_w = 0$, there are two cases to consider:

- If there exists two subspaces with trivial intersection (WLOG, say $\tilde{\ell}_p \cap \tilde{\ell}_q = 0$), then notice that $\tilde{\ell}_p \oplus \tilde{\ell}_q = k^4$ (which is the same as the prove of the **Converse** part of **Problem 1**), then notice in that prove, we've shown that it reaches a contradiction eventually if given that the line $\{tp + rq \mid [t, r] \in \mathbb{P}^1\}$ is contained in $\mathbb{G}$; and since this line is also parametrized by $\{sp + tq + 0w \mid [s, t, 0] \in \mathbb{P}^2\} \subset \Lambda$, which is assumed to be contained in $\mathbb{G}$, then here it reaches a contradiction also.
- Else, if every two subspaces have nontrivial intersection, then since $\tilde{\ell}_p \cap \tilde{\ell}_q \cap \tilde{\ell}_w = 0$, then one gets nonzero vectors $x \in \tilde{\ell}_p \cap \tilde{\ell}_q$ (while $x \notin \tilde{\ell}_w$), $y \in \tilde{\ell}_p \cap \tilde{\ell}_w$ (while $y \notin \tilde{\ell}_q$), and $z \in \tilde{\ell}_q \cap \tilde{\ell}_w$ (while $z \notin \tilde{\ell}_p$). So, this enforces the list x, y, z to be linearly independent, and $\tilde{\ell}_p = \text{span}\{x, y\}$, $\tilde{\ell}_q = \text{span}\{x, z\}$, and $\tilde{\ell}_w = \text{span}\{y, z\}$. But now, notice that let $\tilde{H} = \text{span}\{x, y, z\}$ be a 3-dimensional subspace of k^4 (so $H \subset \mathbb{P}^3$ is a corresponding plane), we have $\tilde{\ell}_p, \tilde{\ell}_q, \tilde{\ell}_w \in \Sigma_H$. If extend the basis to $a, x, y, z \in k^4$, then under this basis, we get the following matrix representatives:

$$\ell_p = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad \ell_q = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \ell_w = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (4.6)$$

Where, under this Plücker Embedding, $p = P(\ell_p) = [0, 0, 0, 1, 0, 0]$, $q = P(\ell_q) = [0, 0, 0, 0, 1, 0]$, and $w = P(\ell_w) = [0, 0, 0, 0, 0, 1]$. Which, given plane embedding $\varphi : \mathbb{P}^2 \rightarrow \mathbb{P}^5$ by $\varphi([s, t, r]) = [0, 0, 0, s, t, r]$, this implies that $P(\Sigma_H) \subseteq \varphi(\mathbb{P}^2)$. But, the other containment $P(\Sigma_H) \supseteq \varphi(\mathbb{P}^2)$ is also true due to the proof of part **2** of this problem. Hence, we deduce that $\varphi(\mathbb{P}^2) = \{[0, 0, 0, s, t, r] \mid [s, t, r] \in \mathbb{P}^2\} = \{sp + tq + rw \mid [s, t, r] \in \mathbb{P}^2\} = \Lambda$, is characterized by embedding of Σ_H . However, this again contradicts our initial assumption that $\Lambda \neq \Sigma_H$ for any plane $H \subset \mathbb{P}^3$.

Since in both cases we get a contradiction, the assumption must be false, hence $\tilde{\ell}_p \cap \tilde{\ell}_q \cap \tilde{\ell}_w \neq 0$, showing there exists nonzero $x \in \tilde{\ell}_p \cap \tilde{\ell}_q \cap \tilde{\ell}_w$. Which, this shows that $\ell_p, \ell_q, \ell_w \in \Sigma_x$ (the lines passing through $x \in \mathbb{P}^3$), which reduces to the case in part **1** of this problem, which shows that $\Lambda =$

■ Σ_x . So, we conclude that if plane $\Lambda \subset \mathbb{G}$ has $\Lambda \cong \mathbb{P}^2$, then it must be either Σ_H for some plane $H \subset \mathbb{P}^3$, or Σ_p for some point $p \in \mathbb{P}^3$.

5 D

Problem 3

Let $\ell_1, \ell_2 \subset \mathbb{P}^3$ be disjoint lines. Show that the set $Q \subset \mathbb{G}$ of lines in $\mathbb{P}^3$ meeting both is the intersection of $\mathbb{G}$ with a 3-plane $\mathbb{P}^3 \subset \mathbb{P}^5$, and so is a quadric surface. Deduce from this that $Q \cong \mathbb{P}^1 \times \mathbb{P}^1$.

Solution: Given that $\ell_1 \cap \ell_2 = \emptyset$, this implies that their corresponding 2-dimensional subspace $\tilde{\ell}_1, \tilde{\ell}_2 \subset k^4$ has $\tilde{\ell}_1 \cap \tilde{\ell}_2 = 0$, which further implies that $\tilde{\ell}_1 \oplus \tilde{\ell}_2 = k^4$ (since again $\dim_k(\tilde{\ell}_1 \oplus \tilde{\ell}_2) = \dim_k(\tilde{\ell}_1) + \dim_k(\tilde{\ell}_2) = 4$). Which, choose basis $e_0, e_1 \in \tilde{\ell}_1$ and $e_2, e_3 \in \tilde{\ell}_2$, the list $e_0, e_1, e_2, e_3 \in k^4$ forms a basis, which we'll use it for the corresponding Plücker Embedding.

Now, given any line $\ell \in Q$, since $\ell \cap \ell_1$ and $\ell \cap \ell_2$ are both nontrivial, then its corresponding 2-dimensional subspace $\tilde{\ell} \subset k^4$ satisfies $\tilde{\ell} \cap \tilde{\ell}_1, \tilde{\ell} \cap \tilde{\ell}_2 \neq 0$, hence one can choose nonzero vector $x \in \tilde{\ell} \cap \tilde{\ell}_1$ (where $x = x_0 e_0 + x_1 e_1$), and $y \in \tilde{\ell} \cap \tilde{\ell}_2$ (where $y = y_2 e_2 + y_3 e_3$). Under the ordered basis $e_0, e_1, e_2, e_3 \in k^4$, ℓ has the following matrix representative:

$$\ell = \begin{pmatrix} x_0 & x_1 & 0 & 0 \\ 0 & 0 & y_2 & y_3 \end{pmatrix} \quad (5.1)$$

Hence, under the Plücker Embedding, it yields the following:

$$P(\ell) = [0, x_0 y_2, x_0 y_3, x_1 y_2, x_1 y_3, 0] \quad (5.2)$$

Which, if consider the hyperplane embedding $\varphi : \mathbb{P}^3 \rightarrow \mathbb{P}^5$ by $\varphi([s, t, r, w]) = [0, s, t, r, w, 0]$, then the above implies that $P(Q) \subseteq \varphi(\mathbb{P}^3)$, or $P(Q) \subseteq \mathbb{G} \cap \varphi(\mathbb{P}^3)$.

Also, to show that $\mathbb{G} \cap \varphi(\mathbb{P}^3) = P(Q)$, if given any $[0, s, t, r, w, 0] \in \mathbb{G} \cap \varphi(\mathbb{P}^3)$, there are two cases to consider:

- If $s \neq 0$ (WLOG, assume $s = 1$), then if $\ell \in Q$ has $P(\ell) = [0, s, t, r, w, 0]$, one has $x_0 y_2 = s = 1$, so WLOG assume $x_0 = y_2 = 1$ (since it's up to scalar multiplication of the vectors). Hence, we yield:

$$[0, s, t, r, w, 0] = P(\ell) = [0, 1, y_3, x_1, x_1 y_3, 0] \quad (5.3)$$

However, recall that in lecture we've covered that image of Plücker Embedding satisfies the equation $X_{01}X_{23} - X_{02}X_{13} + X_{03}X_{12} = 0$ (**Rmk:** In the lecture it's using $\{1, 2, 3, 4\}$ instead of $\{0, 1, 2, 3\}$ as indices, I accidentally shifted the indices when starting this exam...). Here for point $[0, s, t, r, w, 0] \in \mathbb{G}$, $X_{01} = 0, X_{02} = s, X_{03} = t, X_{12} = r, X_{13} = w, X_{23} = 0$, hence we get $-sw + tr = 0$; and with assumption $s = 1$, we yield $w = tr$. Hence, for $x_1 = r$ and $y_3 = t$ (also $x_0 = y_2 = 1$), it satisfies $\ell = \begin{pmatrix} 1 & r & 0 & 0 \\ 0 & 0 & 1 & t \end{pmatrix}$ and $P(\ell) = [0, 1, t, r, tr, 0] = [0, s, t, r, w, 0]$, showing this point is contained in $P(Q)$.

- Else, if $s = 0$, then if $\ell \in Q$ has $P(\ell) = [0, s, t, r, w, 0]$, one has $x_0 y_2 = s = 0$. This generates two extra cases:
 - If $r = 0$, choose $y_2 = 0$. Which, we yield:

$$[0, s, t, r, w, 0] = P(\ell) = [0, 0, x_0 y_3, 0, x_1 y_3, 0] \quad (5.4)$$

Under this case, for $P(\ell)$ to be well-defined, one must have either $x_0 y_3$ or $x_1 y_3 \neq 0$, showing $y_3 \neq 0$ (if $y_3 = 0$, $x_0 y_3 = 0 = x_1 y_3$ violates the definition that $P(\ell) \in \mathbb{P}^5$). Hence, can assume $y_3 = 1$. Then, we yield that $x_0 = t$ and $x_1 = w$. Which, $\ell = \begin{pmatrix} t & w & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$, or $P(\ell) = [0, 0, t, 0, w, 0] = [0, s, t, r, w, 0]$.

• Else if $r \neq 0$, choose $x_0 = 0$ instead. Which, we yield:

$$[0, s, t, r, w, 0] = P(\ell) = [0, 0, 0, x_1 y_2, x_1 y_3, 0] \quad (5.5)$$

For the point $P(\ell) \in \mathbb{P}^5$, one must have one of $x_1 y_2$ or $x_1 y_3$ being nonzero, showing $x_1 \neq 0$. Hence, can assume $x_1 = 1$. Then, we yield that $y_2 = r$ and $y_3 = w$. Which, $\ell = \begin{pmatrix} x_0 & x_1 & 0 & 0 \\ 0 & 0 & r & w \end{pmatrix}$, or $P(\ell) = [0, 0, 0, r, w, 0] = [0, s, t, r, w, 0]$ (**Rmk:** The equation for the image of P shows that $X_{01}X_{23} - X_{02}X_{13} + X_{03}X_{12} = 0$, so $-sw + tr = 0$. With assumption $s = 0$, then $tr = 0$; and with assumption $r \neq 0$, we must have $t = 0$).

Hence, in both casees for $s = 0$, one still has $[0, s, t, r, w, 0] \in P(Q)$.

So, regardless of the case we always have $\mathbb{G} \cap \varphi(\mathbb{P}^3) \subseteq P(Q)$, which shows that $P(Q) = \mathbb{G} \cap \varphi(\mathbb{P}^3)$. Hence, Q can be characterized as $\mathbb{G}$ intersecting $\mathbb{P}^3$, which is a quadric surface.

Finally, Since based on the basis we chose, we have every $\ell \in Q$ being characterized as $\ell = \begin{pmatrix} x_0 & x_1 & 0 & 0 \\ 0 & 0 & y_2 & y_3 \end{pmatrix}$ for some $[x_0, x_1], [y_2, y_3] \in \mathbb{P}^1$ (since one needs $x = x_0 e_0 + x_1 e_1, y = y_2 e_2 + y_3 e_3$ to both be nonzero, in case for $\tilde{\ell} = \text{span}\{x, y\}$ being 2-dimensional). Then, if consider the Segre Embedding $S : \mathbb{P}^1 \times \mathbb{P}^1 \rightarrow \mathbb{P}^3$ by $S([x_0, x_1], [x_2, x_3]) = [x_0 x_2, x_0 x_3, x_1 x_2, x_1 x_3]$, then its image under the embedding $\varphi : \mathbb{P}^3 \rightarrow \mathbb{P}^5$ is given by:

$$\varphi \circ S([x_0, x_1], [x_2, x_3]) = \varphi([x_0 y_2, x_0 y_3, x_1 y_2, x_1 y_3]) = [0, x_0 y_2, x_0 y_3, x_1 y_2, x_1 y_3, 0] \quad (5.6)$$

Which, it is precisely described by $P(\ell)$, given that $\ell = \begin{pmatrix} x_0 & x_1 & 0 & 0 \\ 0 & 0 & y_2 & y_3 \end{pmatrix}$. This shows that $Q \cong \mathbb{P}^1 \times \mathbb{P}^1$ (since $P(Q) = \varphi \circ S(\mathbb{P}^1 \times \mathbb{P}^1)$).